A Reinforcement Learning Framework for Scattering Feature Extraction and SAR Image Interpretation

Jiacheng Chen, *Student Member, IEEE*, Xu Zhang, *Graduate Student Member, IEEE*, Haipeng Wang, *Senior Member, IEEE*, and Feng Xu, *Senior Member, IEEE*

***Abstract*— With the rapid development and widespread deployment of radar technology, the interpretation of the vast amount of synthetic aperture radar (SAR) imagery obtained daily has emerged as a hot topic. The complexity of the electromagnetic scattering mechanisms contained within radar images makes SAR image interpretation a challenging task. Current methodologies for SAR image interpretation primarily involve feature extraction-based techniques, categorized into image-domain and frequency-domain algorithms. However, these methods are faced with issues, such as rough segmentation in images, high-computational complexity, and poor robustness, presenting significant challenges in the field. In this article, a novel framework for SAR image interpretation is proposed, leveraging reinforcement learning (RL) for the extraction of electromagnetic scattering features and the inversion of parameters. Within this framework, a nonsparse reward function, combined with curriculum learning, is introduced as the supervisory information. It enables more efficient policy updates through a structured two-stage training approach. In addition, an algorithm that integrates a four-neighbor breadth-first search (BFS) with the watershed segmentation process is proposed, aiming to enhance the accuracy of scattering center analysis in SAR imagery. The attribute scattering center model (ASCM) is utilized as a prototype for conducting algorithmic research and experimentation. Experimental results on both simulation data and measured data have indicated that the proposed method significantly improves efficiency while ensuring accuracy, demonstrating its capability to extract parameters from measured data in most scenarios.**

***Index Terms*— Attributed scattering center (ASC), machine learning, parameters retrieval, reinforcement learning (RL).**

I. Introduction

A SYNTHETIC aperture radar (SAR) is a microwave remote sensing technology that employs electromagnetic waves as the carrier. Esteemed for its all-weather, all-day functionality, and extensive long-range capabilities, SAR has been extensively integrated into a wide range of military and civilian applications [1]. Advances in radar technology have significantly boosted SAR's capability to produce high-resolution images, thereby revealing more intricate details of the observed targets. Currently, the widespread deployment of radar systems has led to a dramatic increase in data generation. Nevertheless, the inherent complexity of radar imagery, primarily due to the complexity of electromagnetic scattering mechanism, presents substantial challenges in manual interpretation. This has spurred the development of more advanced techniques for SAR image interpretation [2], [3], [4], reinforcing its pivotal role in the realms of remote sensing and intelligence analysis.

The electromagnetic scattering characteristics of targets are correlated with geometrical structures and physical parameters, providing crucial support for SAR image interpretation. In recent years, accurate extraction of electromagnetic features and the inversion of target properties using these features have received extensive attention. In general, electromagnetic features include scattering center features and polarization features [5]. Among them, the echoes of radar targets can be considered as coherent summation of electromagnetic scattering from multiple local scattering sources, which are referred to as scattering centers [6], [7]. The features of scattering centers reflect the geometric attributes of the target, as well as how the echo varies with frequency and azimuth angle. Accurately extracting these scattering center features from SAR echo is crucial. It not only augments our understanding of the target's electromagnetic scattering mechanisms but is also vitally important for the intelligent identification of SAR targets [8], [9].

Currently, research on electromagnetic feature extraction algorithms mainly consists of two categories: image-domain algorithms and frequency-domain algorithms. The image-domain algorithms involve extracting scattering centers from SAR image data within the spatial domain followed by iterative optimization using heuristic algorithms, such as pseudo-Newton methods and segmented patch-based approaches [10], [11], [12], [13]. The ultimate goal of these techniques is to approximate the maximum likelihood solution [14], [15]. However, these algorithms often face challenges, such as cumulative errors stemming from rough segmentation [16], [17], leading to unsatisfactory extraction results, low robustness, and the inability to guarantee global

Manuscript received 15 January 2024; revised 6 May 2024; accepted 2 June 2024. Date of publication 17 June 2024; date of current version 27 June 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62271153; in part by the Natural Science Foundation of Shanghai under Grant 22ZR1406700; and in part by the Open Fund of National Key Laboratory of Scattering and Radiation, Shanghai Radio Equipment Research Institute, under Grant 802NKL2023-002. *(Corresponding author: Haipeng Wang.)*

The authors are with the Key Laboratory for Information Science of Electromagnetic Waves (MoE), Fudan University, Shanghai 200433, China (e-mail: jcchen21@m.fudan.edu.cn; xuzhang19@fudan.edu.cn; hpwang@fudan.edu.cn; fengxu@fudan.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3415614

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:52:56 UTC from IEEE Xplore. Restrictions apply.

optimal solution [6]. The second category focuses on exploiting the sparsity of scattering centers within frequency-domain SAR data, employing Bayesian learning and dictionary-based orthogonal matching pursuit techniques [18], [19], [20], [21]. Nonetheless, these methods grapple with large parameter spaces, necessitating repeated optimization and resulting in significant time and space complexity, which constrains their practical application [22], [23], [24].

The extracted electromagnetic features are critical for subsequent tasks, such as SAR image interpretation and target recognition, which typically employ feature and decision fusion methods. However, the current scattering center extraction algorithms, as mentioned, demand considerable computational resources and often lack the capacity for real-time, high-precision solutions. As a result, some intelligent target recognition methods integrating fused electromagnetic features rely on preprocessed data. This reliance indicates that the process of extracting scattering centers has emerged as a bottleneck, impeding the advancement of comprehensive, end-to-end SAR target intelligent recognition algorithms that utilize electromagnetic feature fusion [25], [26], [27], [28], [29], [30].

To this end, this article aims to rapidly perform radar image interpretation based on the rapid and accurate characterization extraction and parameter inversion of scattering centers in SAR images. In particular, the attribute scattering center model (ASCM) is used as a foundational framework for characterizing scattering mechanisms of scattering centers [31]. The process of retrieving ASCM parameters from SAR images is identified as a high-dimensional, nonlinear, and nonconvex challenge in model parameter optimization [15]. Compared with traditional optimization methods and conventional intelligence approaches, reinforcement learning (RL), a sophisticated machine learning approach, has many advantages, such as adaptability, autonomous learning from interaction, lower dependency on data, and enhanced generalization capabilities. Therefore, RL is applied in this study to tackle the complex task of parameters retrieval. It is used to enable agents to learn optimal decision-making through trial and error while interacting with their environment. Specifically, each optimization iteration is treated as a decision-making step by the agent, making the use of RL for parameter extraction a promising field of exploration [32], [33]. The RL learning process involves the following: 1) at each time step, the agent interacts with the environment and receives observations, which are processed via deep learning methods to derive both abstract and concrete state features; 2) the actions are assessed based on expected rewards, with a policy mapping the current state to a respective action; and 3) the environment's response to this action results in the next observation. By repeatedly iterating these steps, an optimal policy that achieves the set goals can ultimately be formulated [34].

The main contributions of this article can be summarized as follows.

1) A novel framework for SAR image interpretation is proposed that leverages RL for the extraction of electromagnetic scattering features and the inversion of parameters. In the RL, parametric models are conceptualized as environments, enabling meaningful interaction with the agent. The ASCM is selected as a prototypical model for conducting algorithmic research and experimentation within this framework. Building upon this innovative approach, this article successfully executes the extraction and parameter inversion of ASC scattering centers. Furthermore, the proximal policy optimization (PPO) [35] algorithm is employed to refine the agent's optimization strategy throughout the training phase, enhancing the efficiency and effectiveness of the model.
2) A nonsparse reward is designed to prevent the issue of sparse rewards during the training process of the agent. As the only supervisory information in RL, rewards need to be nonsparse; otherwise, it is difficult for the policy to be effectively updated. The nonsparse reward designed by this article effectively prevents the problem of sparse reward [36] in combination with curriculum learning [37]; therefore, the agent can get efficient policy learning in the training stage. Curriculum learning involves the strategic decomposition of a complex task into simpler, sequential tasks for stagewise training. For the practical application of this study, a two-stage training approach is adopted, enabling a more structured and effective learning process for the agent. This approach significantly enhances the agent's learning efficiency and policy development in RL settings.
3) A four-neighbor breadth-first search (BFS) algorithm is proposed and integrated into the watershed segmentation process to improve the precision of scattering center segmentation. The algorithm proposed in this article adheres to the established paradigm of image-domain methods, where a critical initial step is the decoupling of scattering centers, which is pivotal for following accurate extraction [6]. Experimental results demonstrate that incorporating the BFS algorithm significantly enhances the efficacy of decoupling scattering centers, outperforming traditional watershed segmentation methods. This advancement underscores the algorithm's improved capability in accurately segmenting and analyzing scattering centers within SAR imagery.

The rest of this article is organized as follows. Section II delineates the modeling and explication of the algorithmic framework presented in this article, elucidating the issues at hand. Then, it introduces the basic knowledge of ASCM and the principles of RL. Section III elucidates the methodologies proposed in this article along with the requisite background knowledge. In Section IV, the experimental results of the proposed method on simulated and measured SAR data are presented. Meanwhile, conventional image-domain and frequency-domain methods are tested for comparative analysis. Finally, the conclusions and future work are drawn in Section V.

II. Problem Modeling

A. Problem Description

In this study, the scattering result is denoted as $\boldsymbol{E}(f, \varphi; \Theta)$, where f is the frequency, φ is the aspect angle, and Θ is the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:52:56 UTC from IEEE Xplore. Restrictions apply.

parameters set of the model. The SAR echo is expressed as a 2-D function $\boldsymbol{X}(f,\varphi)$. The primary goal is to optimize the parameters Θ of the scattering center, aiming to ensure that the simulated result $\boldsymbol{E}$ of the model closely matches the actual SAR echo $\boldsymbol{X}$.

To achieve this, an RL approach is employed to train an agent. In RL settings, the policy network of the agent is denoted as ψ, which plays a crucial role in determining the optimal adjustments of the parameters Θ. One iteration of optimization can be described as follows:

$$\mathcal{A}_{t+1} = \psi(\mathcal{S}_t) \tag{1}$$

$$\Theta_{t+1} = \Theta_t + \mathcal{A}_{t+1} \tag{2}$$

where Θ_t and Θ_{t+1} are the parameter sets at time steps t and $t+1$, respectively. The action $\mathcal{A}_{t+1}$ dictates the adjustment of the parameters. It is decided by the policy network ψ based on the state $\mathcal{S}_t$ of current time step t. In this process, the duration extends over T time steps, which effectively corresponds to T iterations of parameter optimization.

The optimization problem can be mathematically formulated as follows:

$$\min_{\Theta} \ \mathcal{L}(\boldsymbol{E}(f,\varphi;\Theta),\ \boldsymbol{X}(f,\varphi)) \tag{3}$$

where $\mathcal{L}$ is the loss function that quantifies the discrepancy between the simulated result $\boldsymbol{E}(f,\varphi;\Theta)$ and the actual SAR echo $\boldsymbol{X}(f,\varphi)$.

Thus, the problem involves minimizing the loss function L through parameter adjustments Θ, guided by the RL policy network ψ. The absolute error is employed as the loss function, and (3) can be expressed as follows:

$$\psi^* = \arg\min_{\psi} \left| \boldsymbol{X}(f,\varphi) - \boldsymbol{E}\left(f,\varphi|\Theta_{\boldsymbol{0}} + \sum_{t=1}^{T} \psi(s_t) \right) \right|. \tag{4}$$

B. Attribute Scattering Center Model

The ASCM is a 2-D scattering center model of the target based on geometric diffraction theory and physical optical solutions of canonical objects [38], [39], [40]. Its parameters contain rich physical and geometric properties of the scattering center. The ASCM is expressed as (5) [39]

$$\boldsymbol{E}(f,\varphi;\boldsymbol{\Theta}) = \sum_{i=1}^{N} \boldsymbol{E}_i(f,\varphi;\boldsymbol{\Theta}_i) \tag{5}$$

$$\boldsymbol{\Theta}^{\mathrm{T}} = \left[\boldsymbol{\Theta}_1^{\mathrm{T}}, \boldsymbol{\Theta}_2^{\mathrm{T}}, \ldots, \boldsymbol{\Theta}_N^{\mathrm{T}}\right] \tag{6}$$

in which $\boldsymbol{\Theta}$ is the parameters sets of N scattering centers and $(\cdot)^{\mathrm{T}}$ denotes the transposition operator. $\boldsymbol{E}$ represents the overall scattering echo. $\boldsymbol{E}_i$ is the scattering result of the ith scattering center

$$\begin{aligned}\boldsymbol{E}_i(f,\varphi;\boldsymbol{\Theta}_i) = A_i \cdot & \left(j\frac{f}{f_c}\right)^{\alpha_i} \\ & \cdot \exp\left(-j\frac{4\pi f}{c}(x_i\cos\varphi + y_i\sin\varphi)\right) \\ & \cdot \operatorname{sinc}\left(\frac{2\pi f}{c}L_i\sin\left(\varphi-\bar{\phi}_i\right)\right) \\ & \cdot \exp(-2\pi f\gamma_i\sin\varphi)\end{aligned} \tag{7}$$

TABLE I
CORRESPONDENCE BETWEEN SCATTERING STRUCTURE AND PARAMETERS α AND L

Scattering structure	α	L
Dihedral	1	>0
Trihedral	1	$=0$
Cylinder	0.5	>0
Top-hat	0.5	$=0$
Sphere	0	$=0$
Edge broadside	0	>0
Edge diffraction	-0.5	>0
Corner diffraction	-1	$=0$

where f represents the frequency, φ represents the aspect angle, and c symbolizes the speed of light. f_c is the center frequency. $\boldsymbol{\Theta}_i = [A_i, x_i, y_i, L_i, \bar{\phi}_i, \alpha_i, \gamma_i]$ is the parameters set of the ith scattering center. A_i is the relative amplitude, while x_i and y_i represent the coordinates of the ith scattering center relative to the zero-phase reference point in the range and azimuth directions, respectively. L_i indicates the length of the ith scattering center. When $L_i = 0$, it indicates that the scattering center is a localized scattering center. Otherwise, it is a distributed scattering center. $\bar{\phi}_i$ is the azimuth angle of the ith scattering center and always equals zero if the corresponding scattering center is a localized scattering center. γ_i represents the azimuth dependence of the scattering amplitude of the local scattering center. α_i represents frequency dependence and can be combined with L_i to derive the scattering structure of the ith scattering center. Therefore, parameter α plays an important role in the parameters retrieval. The correspondence between the scattering structure and the parameters (L,α) is shown in Table I [10], [22], [41]. Table I demonstrates that the structure of the target can be ascertained by accurately determining the values of the parameters L and α.

C. Reinforcement Learning

In RL, there are two fundamental roles: the agent and the environment. The goal is to maximize the rewards through the interaction between the agent and the environment. Fig. 1 provides an illustrative representation of the relationship between the agent and the environment. In a given time step, the agent selects an action $\mathcal{A}_t$ based on the current state $\mathcal{S}_t$ of the environment. This action exerts an influence on the environment, inducing a transition to the subsequent state $\mathcal{S}_{t+1}$. Concurrently, the environment furnishes feedback in the form of a reward r_{t+1} to the agent. Analogously, this iterative interaction persists into the subsequent time step, continuing in such a cyclical fashion until the conclusion of an episode. Throughout this process, the agent accumulates samples throughout an episode, denoted as follows:

$$\{\mathcal{S}_0, \mathcal{A}_0, r_1, \mathcal{S}_1, \mathcal{A}_1, r_2, \ldots, \mathcal{S}_{T-1}, \mathcal{A}_{T-1}, r_T, \mathcal{S}_T\}. \tag{8}$$

Among the episodic interaction samples, it is customary to denote the samples at each time step as $(\mathcal{S}_t, \mathcal{A}_t, r_{t+1}, \mathcal{S}_{t+1})$, representing the state transition from time step t to $t+1$,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:52:56 UTC from IEEE Xplore. Restrictions apply.

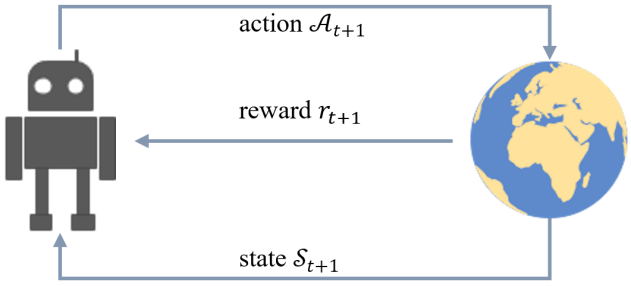


Fig. 1. Fundamental framework of RL.

named "transition." During training, learning is often conducted with these "transitions." One significant reason for this lies in the temporal dependencies present in samples within an episode, which do not adhere to the assumption of each sample being independently and identically distributed (i.i.d.) when using deep neural networks. Consequently, RL employs transition as the training inputs rather than entire episodes. Rewards serve as the sole source of supervisory information in RL training. When we desire the agent to accomplish a specific task, it becomes essential to design rewards that are aligned with the objectives of that task. Simultaneously, it is imperative to ensure that the reward design does not introduce undue bias toward the ultimate learning goal. Assuming that the immediate rewards received at each time step in the future $r(\mathcal{S}_t)$ must be discounted by a discount factor Γ, then the cumulative rewards from time step 0 to time step T, where an episode concludes, are defined as follows:

$$R = \sum_{t=0}^{T} \Gamma^t r(\mathcal{S}_t). \tag{9}$$

The objective of RL is to learn a stochastic policy ψ^*, which is a conditional probability density over actions given the current state, such that the expected cumulative rewards are maximized. That is,

$$\psi^* = \arg\max_{\psi} \mathbb{E}_{\mathcal{S}_0,\mathcal{A}_0,\mathcal{S}_1,\ldots,\mathcal{S}_T}\left[\sum_{t=0}^{T} \Gamma^t r(\mathcal{S}_t)\right]. \tag{10}$$

III. Methodology

The algorithm consists of two stages: the image-domain scattering center decoupling and the parameter optimization. For the input SAR image, the algorithm initially divides it into multiple images, each containing only one scattering center. Subsequently, the agent extracts the parameters separately for these scattering centers. The overall framework of proposed method is illustrated in Fig. 2.

A. BFS Prelabeled Scattering Center Decoupling

The watershed algorithm can be used for the segmentation of scattering centers [6], [10]. However, this algorithm often led to the erroneous segmentation of distributed scattering centers into multiple subscattering centers. The primary reason is the propensity of the watershed algorithm to readily differentiate high-amplitude pixel regions with certain differences within distributed scattering centers into multiple "peaks," therefore resulting in the formation of multiple subscattering center regions during the subsequent labeling process. For image-domain methods, the decoupling of scattering centers in the image domain is of great importance. Once significant errors occur during the decoupling phase, subsequent parameter optimization is adversely affected, resulting in a decrease in accuracy. Therefore, the BFS for prelabeling is introduced in this work, expecting to group high-amplitude pixels with differences within distributed scattering centers into the same "peaks."

First, locate the pixel with peak value $I_{\max}$ as the starting point, and compute a decibel matrix $\mathbf{M} = I_{\max} \cdot 10^{(T_h/10)}$. Then, the algorithm initiates by setting a threshold value T_h and conducting a four-neighborhood traversal. During this traversal, if the decibel value at a given position exceeds the threshold, that position is labeled and added to a queue for storage. The process continues by dequeuing a pixel and repeating the traversal and labeling steps until the queue becomes empty. The traversal sequence is assumed to follow the order of right, up, left, and down. With the threshold T_h set to -3 dB, the labeling procedure of the algorithm is depicted in Fig. 3. Subsequently, the image with preliminary labels is used as the input for the watershed algorithm, which further decouples the scattering centers.

B. Interaction Framework Between Agent and ASCM Forward Process

This section constitutes the core of the proposed method. In the preceding stage, multiple images containing only one scattering center are generated as input for the current stage. Then, the parameters of each scattering center in each image will be sequentially extracted by the agent in the current section.

This interaction framework modeled the form of ASCM forward processing as the RL environment, which interacts with the agent during the parameters optimization. The overall interaction framework is divided into several components for design. First, it is necessary to construct an agent. As described in the previous section, the agent's primary task is to learn a mapping from states to actions that maximize the total reward. In this context, this article employs a policy network to fulfill this mapping responsibility. With the policy network in place, the design of the agent's actions is required. Specifically, this refers to the transformation of the policy network's output to adjustments in the parameters. Second, the appropriate representation of states is of paramount importance. This constitutes the observation information available to the agent and profoundly influences the agent's assessment of the current task conditions, enabling it to execute the correct actions. If the current environment can provide the necessary information to the agent without excessive redundancy, the agent can engage more effectively in policy learning. Third, the reward function needs to be designed to be task guiding. The policy that the agent needs to learn is to understand what action is the right one at a certain time. With the appropriate reward and penalty mechanisms in place, after attempting multiple times, the agent will learn what is the correct policy. The following contents provide a detailed explanation of each component.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:52:56 UTC from IEEE Xplore. Restrictions apply.

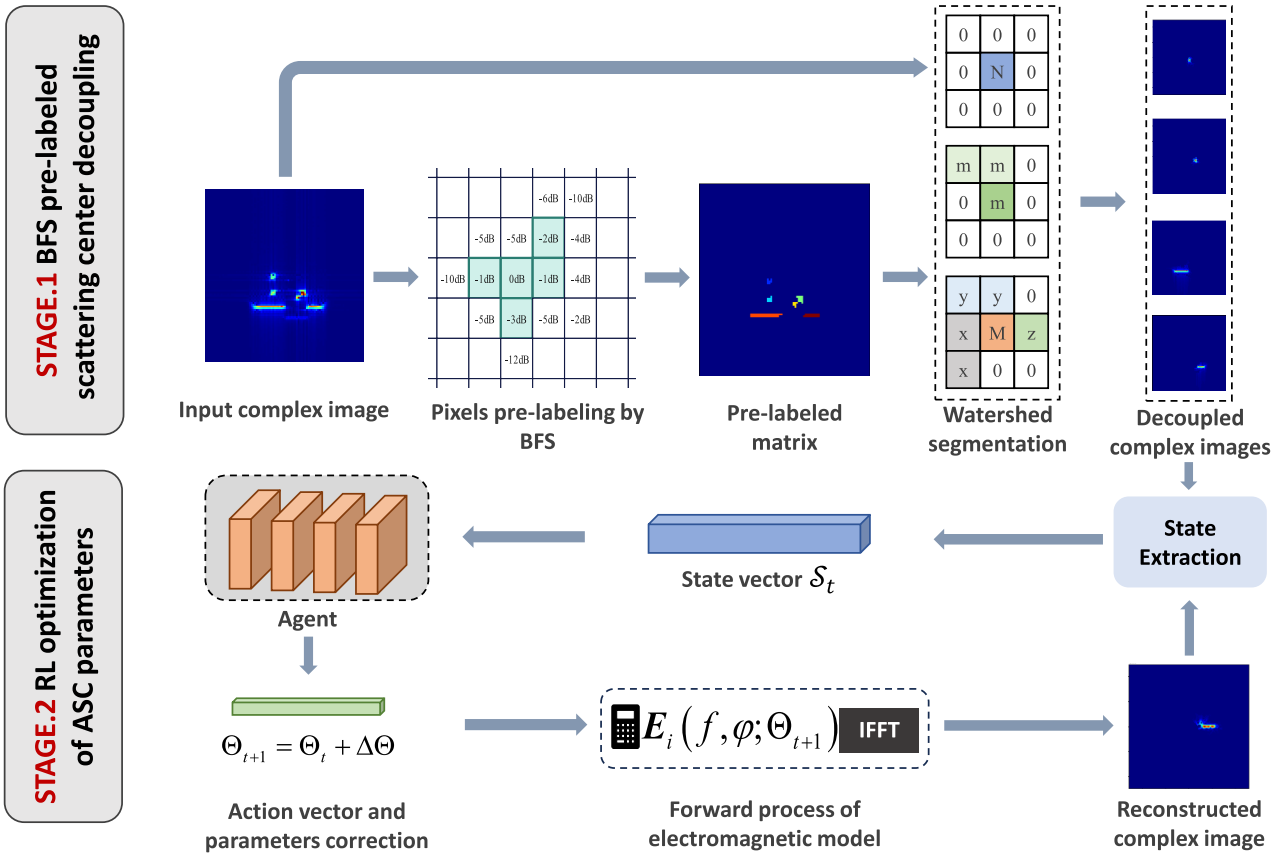


Fig. 2. Overall framework of the proposed method.

	A	B	C	D	E
0				-6dB	-10dB
1		-5dB	-5dB	5 -2dB	-4dB
2	-10dB	3 -1dB	1 0dB	2 -1dB	-4dB
3		-5dB	4 -3dB	-5dB	-2dB
4			-12dB		

Fig. 3. Simple example of BFS labeling: yellow numbers represent the order of labeling, and blue pixels indicate the regions marked in the current example.

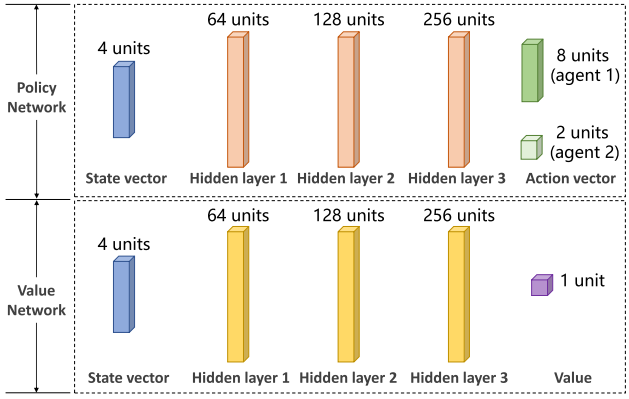


Fig. 4. Architecture of the policy network and value network.

1) Agent Design: The proposed method utilizes the PPO algorithm as the training algorithm for the agent's networks. In the PPO settings, the gradient formula for the policy network also includes an estimate of the value corresponding to each time step in the interaction data. This necessitates the inclusion of two networks in the agent: the policy network and the value network. Both networks take the current state as input, and the policy network outputs actions, while the value network provides a value estimation of the current state. In this context, "value" refers to the expected total reward that the agent can obtain from the current state until the end of an episode. It provides a baseline to measure the advantage of different actions in specific states, which is then used to update the policy network and encourage higher yielding actions. This article opts for a simple multilayer perception (MLP) to construct both the policy network and the value network, and their structures are depicted in Fig. 4.

2) State Extraction: As illustrated in the state extraction module in Fig. 2, the proposed method extracts a state vector that captures the current differences by comparing the input SAR image with the reconstructed SAR image. Intuitively, if the input SAR image closely resembles the reconstructed SAR image, it suggests that the agent has

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:52:56 UTC from IEEE Xplore. Restrictions apply.

optimized parameters quite well. This article represents the state using a vector with a length of 4, denoted as $\mathcal{S} = [s_{\text{Range}}, s_{\text{Azimuth}}, s_L, s_{\text{MAE}}]$. Each element is calculated using the following equations:

$$s_{\text{Range}} = \frac{x_{\max}^{\text{In}} + x_{\min}^{\text{In}}}{2} - \frac{x_{\max}^{\text{Rec}} + x_{\min}^{\text{Rec}}}{2} \tag{11}$$

$$s_{\text{Azimuth}} = \frac{y_{\max}^{\text{In}} + y_{\min}^{\text{In}}}{2} - \frac{y_{\max}^{\text{Rec}} + y_{\min}^{\text{Rec}}}{2} \tag{12}$$

$$s_L = \left(y_{\max}^{\text{In}} - y_{\min}^{\text{In}}\right) - \left(y_{\max}^{\text{Rec}} - y_{\min}^{\text{Rec}}\right) \tag{13}$$

$$s_{\text{MAE}} = \frac{1}{N}\sum_{i=0}^{N-1}(I_i - R_i) \tag{14}$$

where $x_{\max}^{\text{In}}$ and $x_{\min}^{\text{In}}$ are the maximum and minimum values of the x-coordinate among pixels exceed half of the peak value in the input image. $y_{\max}^{\text{In}}$ and $y_{\min}^{\text{In}}$ are the maximum and minimum values of the y-coordinate among pixels exceeding half of the peak value in the input image. The same applies to $x_{\max}^{\text{Rec}}$, $x_{\min}^{\text{Rec}}$, $y_{\min}^{\text{Rec}}$, and $y_{\min}^{\text{Rec}}$, and the only difference is that they come from the reconstructed image. s_L can, to some extent, reflect the length difference between the corresponding scattering centers in the two images, so the first letter L of length is used as the subscript. In (14), I_i and R_i are pixel values in the input image and those in the reconstructed image, respectively.

3) Reward Function: The reward for the proposed method primarily consists of three components: single-step reward r_s, time-step cost r_c, and final reward r_f. r_c is a hyperparameter, and the expressions of others are as follows:

$$r_s = \sum_i |p_{i,t-1} - \hat{p}_i| - \sum_i |p_{i,t} - \hat{p}_i|. \tag{15}$$

$$r_f = \begin{cases} 0, & \varepsilon \le \sum_i \left|p_{i,T} - \hat{p}_i\right| \\ \beta \cdot \left(\varepsilon - \sum_i \left|p_{i,T} - \hat{p}_i\right|\right), & \varepsilon > \sum_i \left|p_{i,T} - \hat{p}_i\right|. \end{cases} \tag{16}$$

The single-step reward r_s provides feedback by measuring the parameter error before and after the agent performs an action. The final reward r_f is given based on the final error at the end of the entire optimization iteration. Among them, $p_{i,t}$ represents the value of the ith parameter at time step t, and $\hat{p}_i$ represents the true value of the ith parameter. β and ε are both hyperparameters. In particular, β is the amplification factor, and ε is the error threshold. The time-step cost r_c incurs a fixed cost during each iteration, encouraging the agent to complete the task as quickly as possible. In addition, to prevent the agent from producing illegal parameter values, a significantly negative reward r_n is returned when the parameters exceed their boundaries. Therefore, the expression for the reward in each iteration t is as follows:

$$r_t = r_s + r_f + r_c + r_n. \tag{17}$$

4) Training and Inference: At each training episode, a set of parameters is randomly generated, and their corresponding reconstructed image plays the role of a real SAR image during training. Based on the interaction framework illustrated in Fig. 2, along with the aforementioned agent network, state extraction, and reward function, the algorithm for training the agent is outlined in Algorithm 1. For each episode, an SAR image is constructed by a set of random parameters $\boldsymbol{\Theta}_i^{\text{gt}}$, reckoned as a truth value. Multiple time steps make one episode. At the first time step, SAR image I_0 is first constructed based on initiated parameters $\boldsymbol{\Theta}_0$. Then, the state vector $\mathcal{S}_0$ is extracted by $s(\cdot)$. An action would be given by the policy network $\boldsymbol{\psi}_{\boldsymbol{w}}$ with the input $\mathcal{S}_0$. The action represents the variation of parameters, i.e., the $\Delta\boldsymbol{\Theta}$. Thereupon, the parameter of next time step $\boldsymbol{\Theta}_1 = \boldsymbol{\Theta}_0 + \Delta\boldsymbol{\Theta}$ is obtained. The reward function can now calculate the reward $\boldsymbol{r}_t$. Before the implementation of the next time step, data in current time step will be collected into the replay buffer $\boldsymbol{D}$. After each episode, a batch of data will be sampled from $\boldsymbol{D}$, and a gradient will be calculated based on the PPO gradient formula to update the weights of the policy network.

Algorithm 1 Training of ASC Parameters Extraction Agent

Require: $s(\boldsymbol{I}_1, \boldsymbol{I}_2)$, $\mathrm{r}(\mathcal{S}, \mathcal{A})$, $\boldsymbol{\psi}_{\boldsymbol{w}}(\mathcal{S})$
  Initiate policy weights: $\boldsymbol{w}$
  for iteration $i = 1, 2, 3, \ldots, N$ **do**
    Initiate parameters: $\boldsymbol{\Theta}_0$
    Randomly initiate ground truth parameters: $\boldsymbol{\Theta}_i^{\text{gt}}$
    $\mathcal{S}_0 = \mathrm{s}(\mathrm{ifft}(\boldsymbol{E}(f, \varphi; \boldsymbol{\Theta}_0)), \mathrm{ifft}(\boldsymbol{E}(f, \varphi; \boldsymbol{\Theta}_i^{\text{gt}})))$
    for time step $t = 1, 2, \ldots, T$ **do**
      $\mathcal{S}_t = \mathrm{s}(\mathrm{ifft}(\boldsymbol{E}(f, \varphi; \boldsymbol{\Theta}_t)), \mathrm{ifft}(\boldsymbol{E}(f, \varphi; \boldsymbol{\Theta}_i^{\text{gt}})))$
      $\boldsymbol{\Theta}_{t+1} \leftarrow \boldsymbol{\Theta}_t + \boldsymbol{\psi}_{\boldsymbol{w}}(\boldsymbol{s}_t)$
      $\boldsymbol{r}_t = \mathrm{r}(\mathcal{S}_t, \boldsymbol{\psi}_{\boldsymbol{w}}(\mathcal{S}_t))$
      Collect $\boldsymbol{d}_t = \{\mathcal{S}_t, \boldsymbol{\psi}_{\boldsymbol{w}}(\mathcal{S}_t), \mathcal{S}_{t+1}, \boldsymbol{r}_t\}$ to replay buffer $\boldsymbol{D}$
    end for
    Sample a batch $\boldsymbol{B}$ from replay buffer $\boldsymbol{D}$
    Compute objective gradient $\Delta R(\boldsymbol{w})$ via PPO [35] using $\boldsymbol{B}$
    $\boldsymbol{w} \leftarrow \boldsymbol{w} + \Delta R(\boldsymbol{w})$
  end for

In the current algorithm implementation, the agent is able to estimate the value of parameters x, y, L, α, and $\bar{\phi}$ through the interaction process. Specifically, two agents are trained for $N = 4000$ episodes to infer parameters. The first agent deduces common parameters x, y, L, and α, while the other deduces the parameter $\bar{\phi}$ if $L > 0$ else $\bar{\phi}$ is set to 0. Considering the small contribution of γ to ASC, the proposed method set default $\gamma = 0$ [22]. The actions corresponding to each parameter include increasing and decreasing a fixed value. Therefore, as shown in Fig. 4, the action vector length of the policy network is set to $4 \times 2 = 8$ and $1 \times 2 = 2$. Finally, the estimated solution of the parameter A can be obtained using the least-squares solution formula. So far, the proposed method has completed the inference of one set of parameters.

As aforementioned, a curriculum learning approach is employed during the training of the agent. Specifically, a two-stage curriculum learning process is implemented. The primary difference between the two stages lies in the sparsity of rewards. In the first stage, the agent is required to reduce the total error of the few parameters it is responsible for to a level below $\varepsilon = 5$ in order to receive a substantial reward, whereas, in the second stage, the agent must reduce

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:52:56 UTC from IEEE Xplore. Restrictions apply.

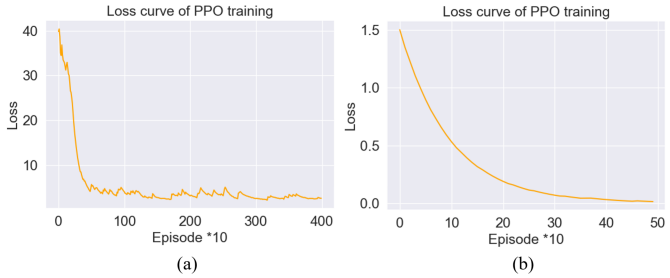


Fig. 5. Loss curve during curriculum training. (a) First stage. (b) Second stage.

this error to below $\varepsilon = 0.5$ to obtain a reward. The training curve for the first agent's training is displayed in Fig. 5. The first stage consisted of 4 000 episodes, while the second stage encompassed 500 episodes. The "loss" represents the final total parameter error at the end of each episode.

IV. Experimental Results

In this section, the experimental analysis begins with tests on reconstructed SAR images. Subsequently, the robustness and noise resistance of the proposed method are evaluated under conditions with different levels of additive white Gaussian noise. The proposed method is further validated with both electromagnetic computation data and measured SAR image chips. Among them, two T72 target SAR image slices from the MSTAR dataset [42] are selected as the measured data. In addition, the OMP method and the RD-AML-CLEAN method were selected as comparative algorithms, and comparative experiments were conducted on reconstruction, simulation, and measured data with the proposed algorithm. To further scrutinize the contributions of this manuscript to the enhancement of algorithm performance, comparative experiments are conducted between algorithms with and without the BFS prelabel module. Simultaneously, ablation experiments are carried out on the various components of the reward function, allowing for an analysis of the roles played by each component of the reward during the process of policy learning by the agent. It is worth noting that the parameter results presented in the experiments are all regularized parameters; hence, the units of the positional parameters and length parameters are in pixels.

Finally, this section delves into the potential and practical significance of integrating the proposed method with intelligent recognition algorithms, focusing on the unique characteristics and advantages highlighted by the experimental results.

A. Test on Reconstructed Data

Reconstructed SAR images are generated randomly, incorporating two localized scattering centers and one distributed scattering center. Assume that the carrier frequency $f_c = 9.6$ GHz and frequency bandwidth $B = 500$ MHz, with the total angular deviation $\phi_m = 3°$ and elevation angle $\theta = 45°$. The number of sampling points in both azimuth and range directions is 100.

Two cases are generated under the settings above. Their ASC ground true values and estimated values given by the proposed method are listed in Table II. The reconstructed original image and the reconstructed image obtained from the estimated values are also shown in Figs. 6 and 7, and each scattering center is marked and corresponds one-to-one with each row of results shown in Table II.

TABLE II
ASC Parameters Extraction Results for Two Reconstructed SAR Images

	No.	Parameters	A	x	y	L	α	$\bar{\phi}$	γ
Case 1	1	Truth	1.34	54.2	40.1	0	1	0	0
		Extracted	1.42	54.7	40.5	0	1	0	0
	2	Truth	15.21	25.0	75.0	21.3	1	0.2	0
		Extracted	16.24	25.3	75.1	21.6	1	0.2	0
	3	Truth	1.21	64.5	43.9	0	0	0	0
		Extracted	1.31	64.0	43.5	0	0	0	0
Case 2	1	Truth	6.36	45.5	41.3	8.3	1	-0.3	0
		Extracted	5.72	44.7	40.9	8.3	1	-0.2	0
	2	Truth	1.53	68.3	49.8	0	0	0	0
		Extracted	1.47	67.7	50.7	0	0	0	0
	3	Truth	1.32	53.8	47.9	0	0.5	0	0
		Extracted	1.34	54.7	48.3	0	0.5	0	0

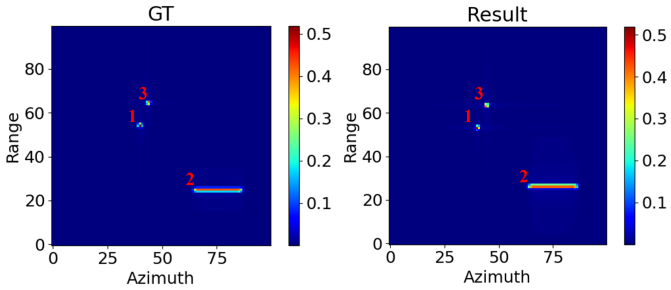


Fig. 6. Case 1: original image and reconstructed image based on estimated ASC parameters using the proposed method.

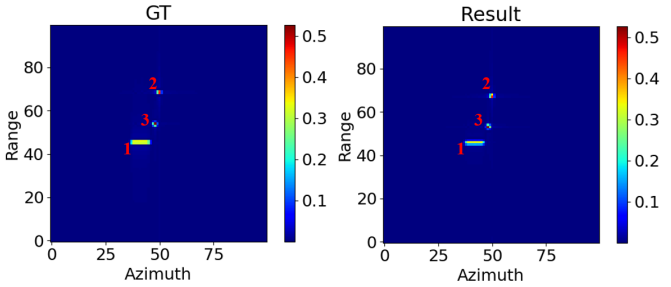


Fig. 7. Case 2: original image and reconstructed image based on estimated ASC parameters using the proposed method.

Furthermore, with the same settings, additive white Gaussian noise is introduced to reconstructed SAR images. Specifically, reconstructed images are generated with the signal-to-noise ratios (SNRs) of 6, 3, 0, −3, and −6 dB, serving as the inputs for the algorithm. Then, the mean absolute error (MAE) of each estimated parameter is calculated under different SNRs. Meanwhile, RD-AML-CLEAN and OMP are also tested for comparison. The result is shown in Fig. 8. It can be found that compared with RD-AML-CLEAN, the proposed method, which is also an image-domain algorithm, has better antinoise performance and accuracy. The OMP method performs the best as a frequency-domain fitting algorithm.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:52:56 UTC from IEEE Xplore. Restrictions apply.

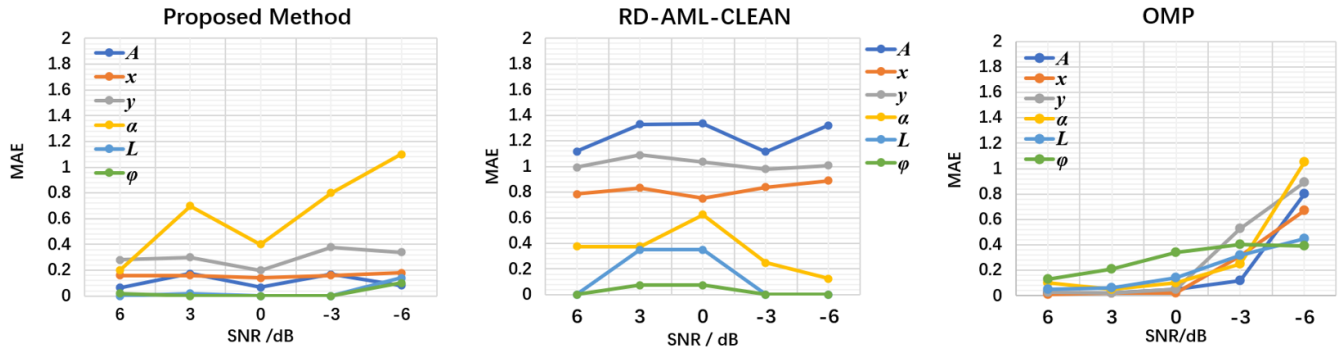


Fig. 8. MAE of each estimated parameter given by the proposed method, RD-AML-CLEAN, and OMP under different SNRs.

TABLE III
AVERAGE MAE AND TIME CONSUMPTION IN 100 RANDOMLY TESTS OF THE PROPOSED METHOD, RD-AML-CLEAN, AND OMP

Method	MAE_A	MAE_x	MAE_y	MAE_α	MAE_L	$\text{MAE}_{\bar{\phi}}$	time consumption (single scattering center)
Proposed Method	0.1	0.16	0.14	0.8	**0.02**	**0.02**	**2.03** s
RD-AML-CLEAN	0.12	0.19	0.09	1	0.06	0.13	203.43 s
OMP	**0.02**	**0.01**	**0.02**	**0.09**	0.03	0.12	154.32 s

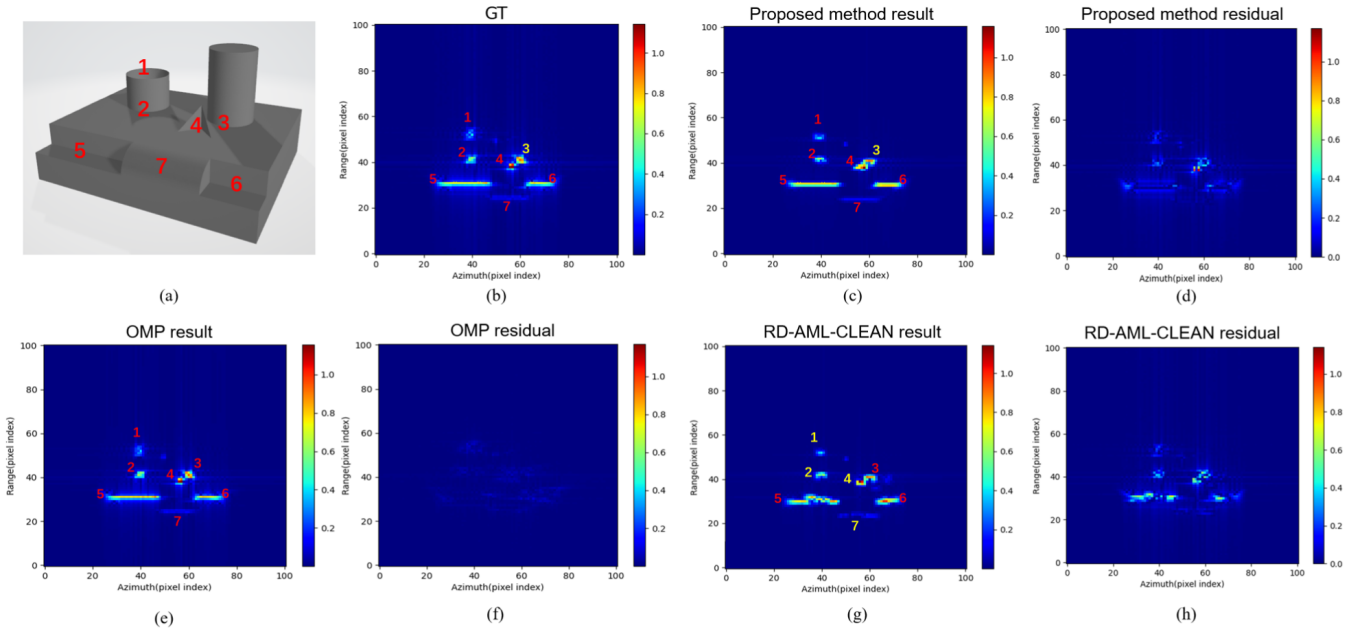


Fig. 9. ASC parameter extraction experiment of the proposed method on SLICY simulation images (structures in which α is incorrectly estimated are marked with yellow numbers): (a) 3-D structure diagram of the SLICY model, (b) SBR electromagnetic simulation SAR image of the SLICY model, (c) reconstructed SAR image of the proposed method, (d) residual of the proposed method, (e) reconstructed SAR image of OMP, (f) residual of OMP, (g) reconstructed SAR image of RD-AML-CLEAN, and (h) residual of RD-AML-CLEAN.

Next, this article compares the proposed method with RD-AML-CLEAN and OMP in terms of accuracy and algorithm time. To avoid the impact of segmentation errors on the image-domain algorithm, 100 SAR images with only one random scattering center are prepared for the comparative experiment. The results are illustrated in Table III. It indicates that the proposed method, while ensuring high accuracy, consumes only approximately 1% of the runtime compared with both RD-AML-CLEAN and OMP methods, significantly improving algorithm efficiency.

The proposed method outperforms RD-AML-CLEAN in terms of accuracy and robustness for several reasons. One is that the former has a better scattering center decoupling algorithm, and the other is that it benefits from manually designed state vector extraction methods. Finally, the decision-making ability of RL algorithms is in unstable states. Compared with the OMP method, although the accuracy of proposed method is not as good as the frequency-domain fitting, it does not require the generation of a dictionary and traversal search, thus saving a lot of time.

B. Test on Simulated Electromagnetic Data

To validate the proposed method, a Sandia laboratories implementation of cylinders (SLICY)-like 3-D model illustrated in Fig. 9(a) is taken as an example to carry out the following simulation. Under a setting of $f_c = 10$ GHz,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:52:56 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
L, α, AND DERIVED STRUCTURES OF EACH SLICY SCATTERING CENTER EXTRACTED BY THE PROPOSED METHOD, OMP, AND RD-AML-CLEAN

Method	Proposed method			OMP			RD-AML-CLEAN		
No.	L	α	Scattering Structure	L	α	Scattering Structure	L	α	Scattering Structure
1	0	0.5	**Top-hat**	4.2	0.5	Cylinder	2	1	Trihedral
2	0	0.5	**Top-hat**	8.05	0.5	Cylinder	3	1	Trihedral
3	0	0	Edge broadside	7.7	0.5	Cylinder	3.4	0.5	Cylinder
4	0	1	**Trihedral**	5.95	1	Dihedral	2.2	0	Sphere
5	21.7	1	**Dihedral**	22.4	1	**Dihedral**	4	1	**Dihedral**
6	9.45	1	**Dihedral**	10.5	1	**Dihedral**	3.8	1	**Dihedral**
7	16.45	0.5	**Cylinder**	16.1	0.5	**Cylinder**	6.6	1	Dihedral

$B = 3$ GHz, $\phi_m = 17°$, and $\theta = 49°$, the high-frequency electrical scattering response of the model is predicted using SBR [43] technology combined with PO and FEC methods. There are 101 sampling points on both azimuth and range directions. The visualized results are shown in Fig. 9, including reconstructed SAR images and residual images of the proposed method, RD-AML-CLEAN, and OMP. From the visualized results, the proposed method extracts the main scattering structures, leaving only the sidelobes around the scattering center. Compared with RD-AML-CLEAN, which is also an image-domain method, it leaves less energy. The OMP method fits data in the frequency domain and can fit almost all energy with additional scattering centers. Therefore, there is virtually no remaining energy.

To facilitate the validation of whether the parameters of each scattering center are correctly extracted for evaluating the algorithm's performance, each scattering center is assigned a unique identifier corresponding to the model's structure. The structure of different scattering centers can be derived by referring to Table I. A threshold is set for L when deriving the scattering structure, and it is considered as $L = 0$ when $L \leq 3$ (corresponding to a local scattering center). The extracted parameters and derived structures of SLICY are listed in Table IV. The correctly inferred scattering structures are shown in bold in the table. The OMP algorithm accurately estimates the frequency-dependent factor α, but it tends to overestimate L, leading to the misidentification of the local scattering center as a distributed scattering center. RD-AML-CLEAN has low accuracy in determining scattering structures due to poor decoupling and inversion precision. The proposed method has slightly lower accuracy in extracting alpha than OMP but overall maintains a good level. Among the seven scattering structures, only one, the Top-hat is misidentified as an edge broadside (No. 3), demonstrating its capability to infer the scattering structures of targets in electromagnetic simulation SAR images. In summary, the proposed method is more suitable for SAR image interpretation tasks, while OMP is more suitable for solving data fitting or image reconstruction tasks.

C. Test on Measured Data

For the measured data, the SAR image of the T72 target contains a clear barrel structure, which is suitable for testing and analyzing scattering feature extraction methods. Thus, two T72 SAR images with different barrel angles are selected for measured data experiment. Considering that there is no true value of the scattering center in the measured image, the algorithms are evaluated from two aspects. One is to calculate the error between the estimated size information of the target and the actual size information derived from multiple estimated scattering center parameters. The other is to observe the visual results between the proposed method and others. The visualized results are shown in Figs. 10 and 11. For size information, three indicators are selected: the body length, width, and barrel length. Combining the pixel resolution of the SAR image, along with position parameters and length parameters derived from algorithms, allows for the actual geometric information of the target to be determined. The corresponding specific results of the proposed method, OMP, and RD-AML-CLEAN are listed in Table V.

It is apparent from the visualized results that the proposed method effectively extracts most of the significant scattering structures of the T72 target, enabling it to gather more comprehensive scattering information compared with the conventional image-domain method RD-AML-CLEAN. The frequency-domain method OMP generates a large number of scattering centers covering the target scattering response, which effectively distinguishes the target from background noise and is suitable for denoising and reconstruction tasks of SAR measured images. When deriving geometric information for the target, the proposed method has higher accuracy compared with RD-AML-CLEAN and is on par with the frequency-domain algorithm OMP, which is known for its high accuracy. Overall, the proposed method can extract the scattering centers of targets in SAR measured images and has the ability to accurately extract target geometric information from SAR images, making it highly applicable for SAR image interpretation tasks.

D. Comparative Experiments Between the Proposed Method With and Without the BFS Prelabeling Module

This section primarily explores the improvement brought about by the introduction of the BFS prelabeling module in the decoupling phase. A total of ten sets of reconstructed data are randomly generated, each comprising three distributed scattering centers and one local scattering center. This setup is chosen, because the effect of BFS prelabeling is mainly evident in segmenting the areas with distributed scattering

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:52:56 UTC from IEEE Xplore. Restrictions apply.

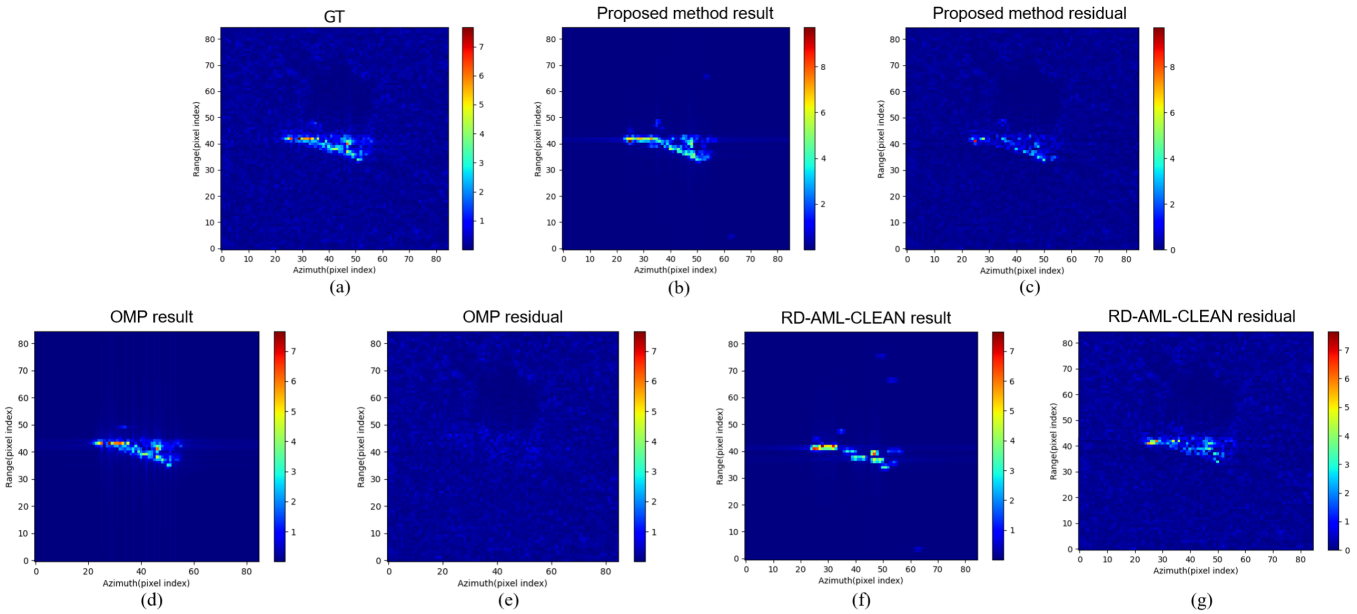


Fig. 10. Visualized results on first T72 measured image: (a) original SAR image, (b) reconstructed SAR image of the proposed method, (c) residual of the proposed method, (d) reconstructed SAR image of OMP, (e) residual of OMP, (f) reconstructed SAR image of RD-AML-CLEAN, and (g) residual of RD-AML-CLEAN.

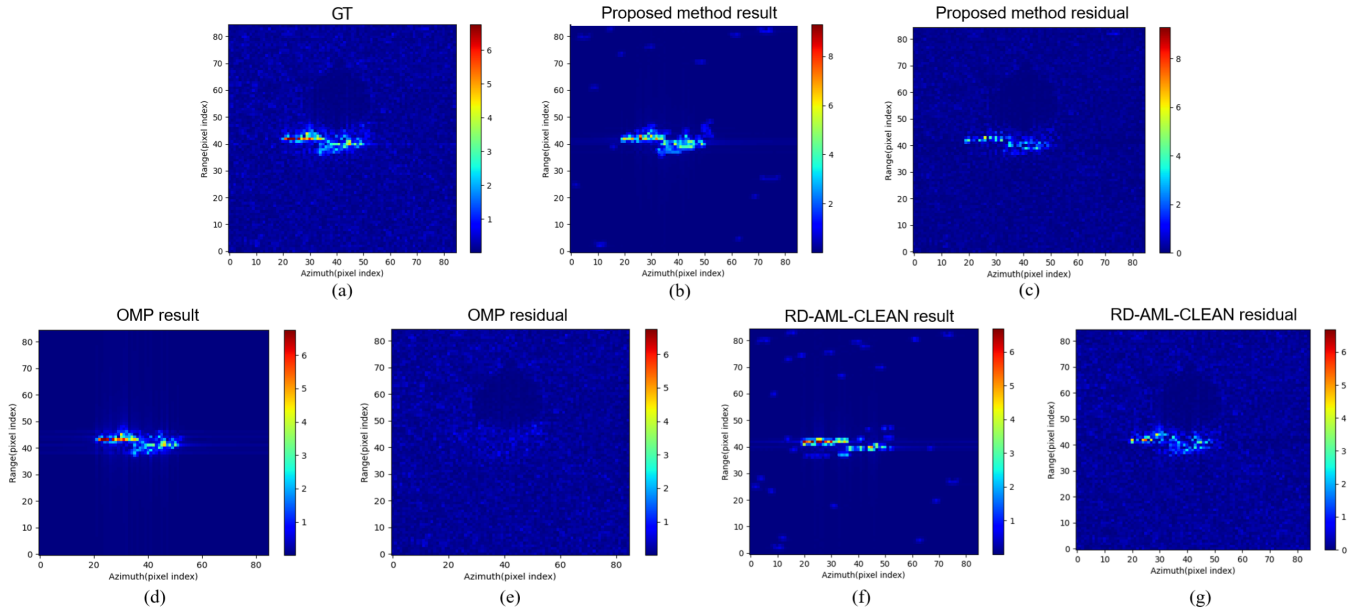


Fig. 11. Visualized results on second T72 measured image: (a) original SAR image, (b) reconstructed SAR image of the proposed method, (c) residual of the proposed method, (d) reconstructed SAR image of OMP, (e) residual of OMP, (f) reconstructed SAR image of RD-AML-CLEAN, and (g) residual of RD-AML-CLEAN.

TABLE V

ESTIMATED SIZE INFORMATION OF T72 USING SCATTERING CENTERS EXTRACTED BY THE PROPOSED METHOD, OMP, AND RD-AML-CLEAN

Parts of T72	Truth	Estimated value of the first T72			Estimated value of the second T72		
		Proposed Method	OMP	RD-AML-CLEAN	Proposed Method	OMP	RD-AML-CLEAN
Barrel length	3.035	**2.999(-0.036)**	2.635(-0.4)	2.71(-0.325)	**3.003(-0.032)**	2.734(-0.301)	3.531(+0.496)
Body width	3.52	3.063(-0.457)	**3.54(+0.02)**	2.798(-0.722)	3.13(-0.39)	**3.619(+0.099)**	3.086(-0.434)
Body length	6.41	**6.228(-0.182)**	6.2(-0.21)	5.925(-0.485)	6.29(-0.12)	**6.377(-0.033)**	7.079(+0.669)

centers. Including a local scattering center is intended to simultaneously test whether BFS prelabeling might impact the accuracy of local scattering center extraction.

Subsequently, the algorithms with and without the BFS prelabeling module are used to extract the scattering center parameters from these data. Then, the number of distributed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:52:56 UTC from IEEE Xplore. Restrictions apply.

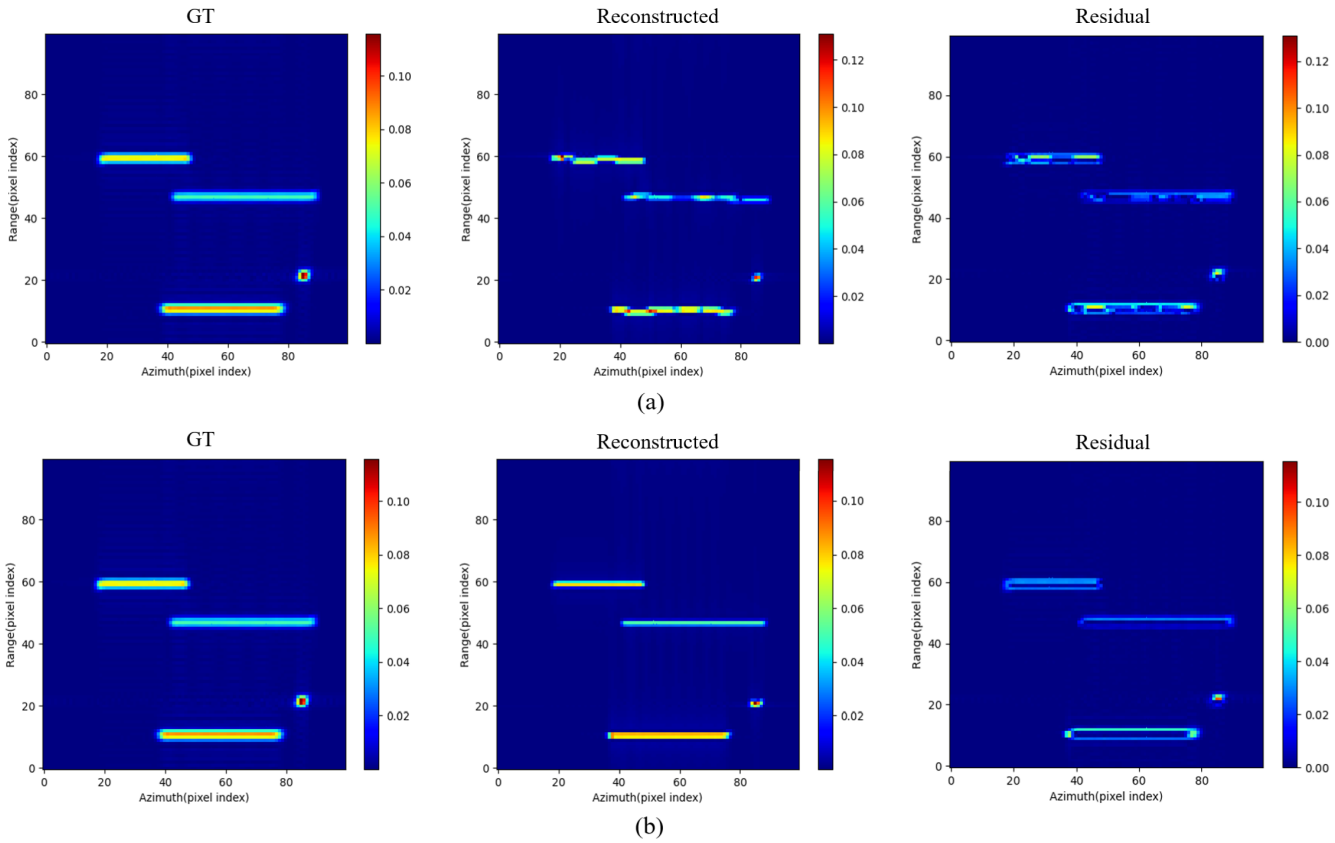


Fig. 12. Example of comparative experiment: (a) result of the proposed method without the BFS prelabeling module and (b) result of the proposed method with the BFS prelabeling module.

TABLE VI
RESULTS OF COMPARATIVE EXPERIMENTS BETWEEN THE PROPOSED METHOD WITH AND WITHOUT THE BFS PRELABELING MODULE

BFS pre-labeling	MAE of local scattering center	Average extracted scattering centers' number	Truth number of scattering centers
✗	1.56	9.6	4
✓	1.49	4	

scattering centers extracted from each dataset and the MAE of the parameters for the local scattering center are calculated for each case. Suppose the number of distributed scattering centers significantly exceeds the true count. In that case, it indicates that the algorithm has erroneously segmented a single distributed scattering center into multiple scattering centers in the decoupling phase. The results are listed in Table VI. To provide a more intuitive illustration of the contrast between the presence and absence of the BFS prelabeling, the visual results are presented in Fig. 12.

From the results, it is evident that the introduction of the BFS prelabeling module allows for a more accurate delineation of scattering center regions. Across the ten randomly generated reconstructed data, the algorithm with the BFS prelabeling module consistently produced four scattering center regions. In contrast, the algorithm without this module tends to erroneously partition distributed scattering centers into multiple subscattering centers, resulting in a greater number of scattering center regions. Furthermore, the results indicate that while BFS prelabeling enhances the decoupling of distributed scattering centers, it does not adversely affect parameter estimation accuracy for local scattering centers.

E. Ablation Experiments for Various Components of Reward Function

To explore the effects of various parts of the reward function design on training, ablation experiments are conducted on each component of the reward. The specific approach of the experiment is to sequentially remove each item of the reward function and then use this reward function to train the agent. The training is set to 4000 episodes, with an interval of discrete action space, which means that the minimum gap of agent parameter tuning is 0.5. Then, draw their loss curves during the training process, which is illustrated in Fig. 13. The loss here refers to the absolute error between the estimated parameters and the actual parameters.

From Fig. 13, it can be observed that the configuration of (14) is the most effective, achieving the lowest loss and the most stable convergence. In contrast, the configuration lacking the term r_s in the reward function failed to converge.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:52:56 UTC from IEEE Xplore. Restrictions apply.

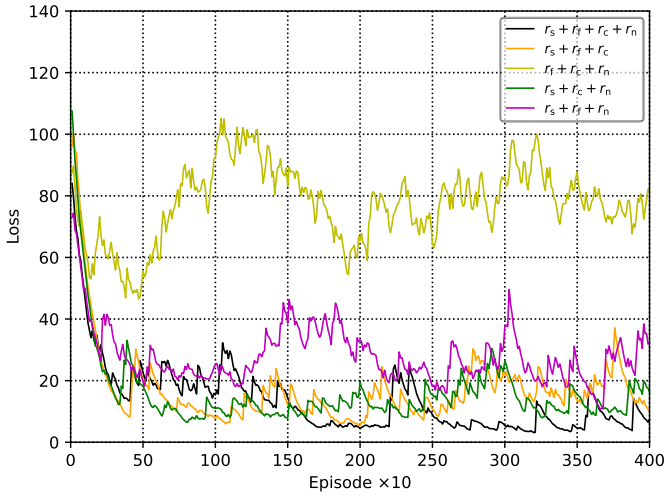


Fig. 13. Loss curves under different reward function configurations.

TABLE VII

PERFORMANCE EVALUATION OF AGENTS TRAINED UNDER DIFFERENT REWARD FUNCTION CONFIGURATIONS

r_s	r_f	r_c	r_n	trainable	MAE	time consumption (per ASC)
✓	✓	✓	✓	✓	2.75	1.97 s
✗	✓	✓	✓	✗	-	-
✓	✗	✓	✓	✓	10.89	2.08 s
✓	✓	✗	✓	✓	15.67	3.21 s
✓	✓	✓	✗	✓	5.67	2.38 s

The main reason is that the absence of r_s creates a problem of sparse rewards during training. The substantial reward r_f only materializes when the parameter error is less than ε as per (16), a condition that is challenging for a random policy to achieve, thus hindering the effective learning of the agent. The configuration without the r_f term can achieve convergence, but in terms of loss and stability, it still falls short of the configuration in (17). The training with the configuration lacking the r_c term can converge, but the loss remains relatively high and exhibits instability. This is mainly because the absence of a soft constraint on the time step length increases the agent's exploration in long time steps during training, while the optimal solution for the current task is clearly not in states with long time steps. The loss curve for the configuration missing the r_n term is similar to that for the configuration without the r_f term. This configuration lacks an explicit constraint on parameter boundaries, resulting in less stable training loss curves.

In addition, for the agents that successfully converged during training, their accuracy and time consumption of extracting each scattering center are tested, and the results are shown in Table VII. It indicates that the reward combination of $r_s + r_f + r_c + r_n$ yields the best-performing agent. r_s serves as the core reward component essential for ensuring successful training, while the other components play supportive roles in enhancing the performance of the agent. It is important to note that the MAE results here may be slightly higher than those shown earlier, because the training of the ablation experiment is only conducted in the first stage.

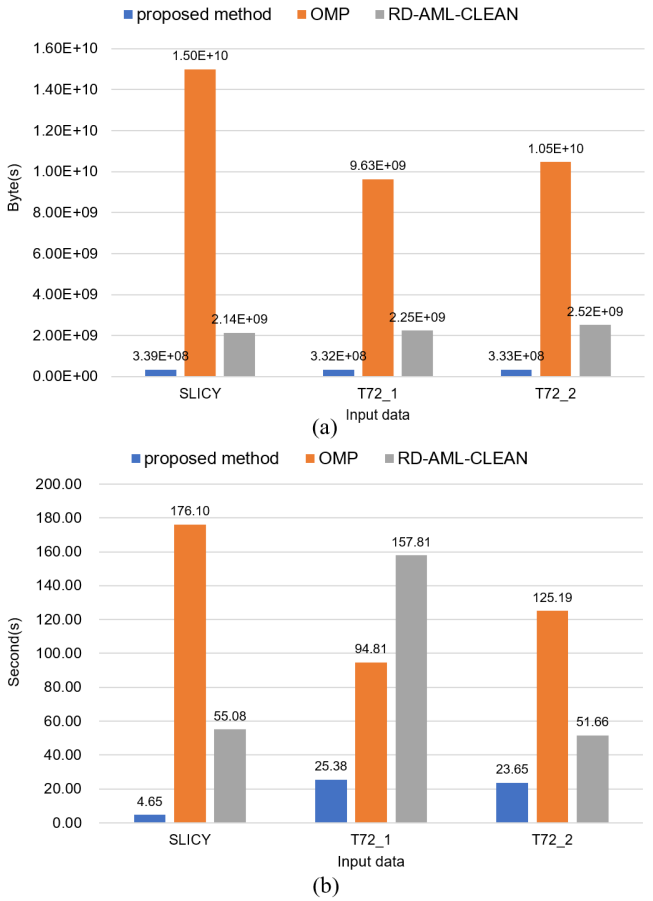


Fig. 14. Memory and times consumption of the proposed method, OMP, and RD-AML-CLEAN: (a) memory consumption bar chart and (b) times consumption bar chart.

F. Memory and Time Consumption

This section compares the proposed algorithm's memory and times consumption with OMP and RD-AML-CLEAN algorithms when processing SLICY simulation data and measured data. The testing hardware environment is a PC with a CPU of i7-10700F, 32 GB of memory, and a graphics card 3060ti. The results of each algorithm are shown in Fig. 14. The proposed method outperforms OMP and RD-AML-CLEAN in both memory resource consumption and time consumption. Among them, the memory consumption is two orders of magnitude smaller than the OMP method and one order of magnitude smaller than the RD-AML-CLEAN method. As an image-domain method, the proposed method's time consumption increases with the number of extracted scattering centers. The algorithm's time consumption is significantly smaller for the number of scattering centers at the target level scale than other methods. Hence, it possesses a distinct advantage in tasks related to interpreting SAR targets.

The efficiency of the proposed method primarily stems from the use of a simple MLP network as the agent's neural network in the current implementation. This choice results in lower gradient backpropagation computational demands and reduced resource requirements. Furthermore, within the framework presented, the agent has already learned strategies for parameter extraction through interaction with the parameterized electromagnetic model. Consequently, in practical

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:52:56 UTC from IEEE Xplore. Restrictions apply.

applications, the well-trained agent can be employed for rapid inference.

Combining the results of the aforementioned experiments, it becomes apparent that the proposed method, in comparison with the existing algorithms, not only ensures high precision but also significantly reduces the algorithm's time consumption. The task of extracting ASC parameters from SAR target image slices has been reduced to a matter of seconds. This substantial improvement enhances the potential application of the proposed method in the intelligent interpretation of SAR images. For instance, in the context of automatic target recognition (ATR) in SAR images, there are already some fusion recognition algorithms utilizing scattering center's features and extracted by deep neural network features [29], [44]. However, these methods typically require the ASC parameters of the SAR image to be prepared in advance. The proposed method, on the other hand, circumvents this inconvenience, making it possible to develop near real-time end-to-end SAR images ATR algorithms that integrate the scattering center's features with image-depth features.

Certainly, the proposed method does have its limitations. First, the algorithm employs discrete actions for parameter adjustment. This becomes a fixed system error, although the training methods of course learning can reduce the system error to a very small extent. Second, the current state vector is manually crafted, and if applied to a similar but different problem, a redesign of the state vector extraction process would be necessary. In addition, the current artificially designed state representation accuracy does not reach the subpixel level, and for sufficiently small errors, the state vector may not be sensitive enough to represent them. In addition, as an algorithm with RL as its core technology, the proposed method requires a large number of interactive iterations to form a reliable policy. This means that each iteration should ideally consume minimal time and resources. Hence, an essential prerequisite for efficiently addressing inverse problems like ASC parameter extraction with such methods is that the computation process of the forward model should be quick and resource-efficient.

V. Conclusion

In this article, a framework utilizing RL for extracting electromagnetic scattering features and inverting parameters is proposed for the first time. The ASCM is chosen as a representative parameterized electromagnetic model and is applied within this framework to conduct algorithmic research and experimental validation. The proposed method follows the paradigm of image-domain algorithms and utilizes BFS prelabeling to improve the performance of scattering center decoupling. The agent training of the proposed method does not require a dataset but is completed through multiple interactions with the forward computation of a parameterized electromagnetic model through randomly generated parameters. Under the nonsparse reward function described in this article, parameter extraction agents can be effectively trained. The experimental results show that the proposed method significantly reduces the time consumption of parameters retrieval, while the accuracy is guaranteed and is able to extract ASC parameters from measured data effectively.

Considering the limitations and the algorithm's application potential, future work can be conducted from the following aspects.

1) An attempt could be made to adopt a continuous action space in order to eliminate the existing system errors. Besides that, an end-to-end state extraction method can be utilized to improve the robustness and applicability of the proposed method.
2) One type of target contains scattering structures that conform to a certain distribution, which is one of the important physical prior information. Works, such as SAR target-level ATR and SAR scene-level 3-D reconstruction, can be further carried out using methods, such as statistics, deep learning, and fusion recognition, based on the distribution of scattering structures provided by the proposed method.
3) Replacing the ASCM with a more sophisticated electromagnetic physics model that possesses enhanced representational capabilities. The primitive scatterer dictionary (PSD) model [7] is a high-frequency scattering modeling method that provides a more specific representation of the target's structures using geometric primitives. Correspondingly, if a similar approach can be used to extract PSD parameters, more detailed and systematic target interpretation information can be obtained. Moreover, it can further validate the broad applicability of the framework proposed in this article.

References

[1] M. Xing, Y. Xie, Y. Gao, J. Zhang, J. Liu, and Z. Wu, "Electromagnetic scattering characteristic extraction and imaging recognition algorithm: A review," *J. Radars*, vol. 11, no. 6, pp. 921–942, 2022.

[2] D. L. Mensa, *High Resolution Radar Imaging*. Dedham, MA, USA: Artech House, 1981.

[3] W. C. Carrara, R. Goodman, R. Majewski, and B. Williams, *Spotlight Synthetic Aperture Radar: Signal Processing Algorithms*. Boston, MA, USA: Artech House, 1995.

[4] J. Jakowatz, D. E. Wahl, P. H. Eichel, D. C. Ghiglia, and P. A. Thompson, *Spotlight-Mode Synthetic Aperture Radar: A Signal Processing Approach*. Norwell, MA, USA: Kluwer, 1996.

[5] A. Moreira, P. Prats-Iraola, M. Younis, G. Krieger, I. Hajnsek, and K. P. Papathanassiou, "A tutorial on synthetic aperture radar," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 1, pp. 6–43, Mar. 2013.

[6] A. Zhang, "Attributed scattering center feature extraction of complex target from high resolution SAR imagery," M.S. thesis, School Electron. Sci. Eng., Nat. Univ. Defense Technol., Changsha, China, 2009.

[7] X. Zhang, F. Xu, Y. Yang, M. Xing, and Y.-Q. Jin, "A primitive scatterer dictionary for semantic representation of radar target images," *IEEE Trans. Antennas Propag.*, vol. 72, no. 1, pp. 825–840, Jan. 2024.

[8] C. Ma, G. Wen, B. Ding, J. Zhong, and X. Yang, "Three-dimensional electromagnetic model–based scattering center matching method for synthetic aperture radar automatic target recognition by combining spatial and attributed information," *J. Appl. Remote Sens.*, vol. 10, no. 1, Mar. 2016, Art. no. 016025.

[9] H.-C. Chiang, R. L. Moses, and L. C. Potter, "Model-based Bayesian feature matching with application to synthetic aperture radar target recognition," *Pattern Recognit.*, vol. 34, no. 8, pp. 1539–1553, Aug. 2001.

[10] M. A. Koets and R. L. Moses, "Image domain feature extraction from synthetic aperture imagery," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, vol. 4, Mar. 1999, pp. 2319–2322.

[11] J. Stach and E. Lebaron, "Enhanced image editing by peak region segmentation," in *Proc. 18th Annu. Meeting Symp. Antenna Meas. Techn. Assoc.*, 1996, pp. 303–307.

[12] M. A. Koets, "Automated algorithms for extraction of physically relevant features from synthetic aperture radar imagery," Ph.D. dissertation, Dept. Elect. Eng., Ohio State Univ., Columbus, OH, USA, 1998.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:52:56 UTC from IEEE Xplore. Restrictions apply.

[13] R. L. Moses, L. C. Potter, and I. J. Gupta, "Feature extraction using attributed scattering center models for model-based automatic target recognition (ATR)," Air Force Research Laboratory (AFRL) and Defense Advanced Research Projects Agency (DARPA), Wright-Patterson AFB, OH, USA, Tech. Rep., AFRL-SN-WP-TR-2006-1004, 2005.

[14] B. Ding and G. Wen, "Target reconstruction based on 3-D scattering center model for robust SAR ATR," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 7, pp. 3772–3785, Jul. 2018.

[15] M. Jing and G. Zhang, "Attributed scattering center extraction with genetic algorithm," *IEEE Trans. Antennas Propag.*, vol. 69, no. 5, pp. 2810–2819, May 2021.

[16] B. G. Clark, "An efficient implementation of the algorithm 'CLEAN,"' *Astron. Astrophys.*, vol. 89, no. 3, p. 377, Sep. 1980.

[17] D. G. Steer, P. E. Dewdney, and M. R. Ito, "Enhancements to the deconvolution algorithm 'CLEAN,"' *Astron. Astrophys.*, vol. 137, no. 2, pp. 159–165, 1984.

[18] X.-Y. Xing, H. Yan, H.-C. Yin, and C.-Y. Huo, "A bistatic attributed scattering center model for SAR ATR," *IEEE Trans. Antennas Propag.*, vol. 69, no. 11, pp. 7855–7866, Nov. 2021.

[19] Y. Xie, M. Xing, Y. Gao, Z. Wu, G.-C. Sun, and L. Guo, "Attributed scattering center extraction method for microwave photonic signals using DSM-PMM-regularized optimization," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022, Art. no. 5230016.

[20] Y. Cong, B. Chen, H. Liu, and B. Jiu, "Nonparametric Bayesian attributed scattering center extraction for synthetic aperture radar targets," *IEEE Trans. Signal Process.*, vol. 64, no. 18, pp. 4723–4736, Sep. 2016.

[21] G. B. Hammond and J. A. Jackson, "SAR canonical feature extraction using molecule dictionaries," in *Proc. IEEE Radar Conf.*, Ottawa, ONT, Canada, Apr. 2013, pp. 1–6.

[22] H. Liu, B. Jiu, F. Li, and Y. Wang, "Attributed scattering center extraction algorithm based on sparse representation with dictionary refinement," *IEEE Trans. Antennas Propag.*, vol. 65, no. 5, pp. 2604–2614, May 2017.

[23] Z. Sun, "Target scattering characteristic extraction method for radar 2-D image in optical region," Ph.D. dissertation, ATR Lab., National Univ Defense Technol., Changsha, China, 2001.

[24] Z. Li, K. Jin, B. Xu, W. Zhou, and J. Yang, "An improved attributed scattering model optimized by incremental sparse Bayesian learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 5, pp. 2973–2987, May 2016.

[25] B. Ding, G. Wen, C. Ma, and X. Yang, "An efficient and robust framework for SAR target recognition by hierarchically fusing global and local features," *IEEE Trans. Image Process.*, vol. 27, no. 12, pp. 5983–5995, Dec. 2018.

[26] T. Li and L. Du, "SAR automatic target recognition based on attribute scattering center model and discriminative dictionary learning," *IEEE Sensors J.*, vol. 19, no. 12, pp. 4598–4611, Jun. 2019.

[27] Y. Li, L. Du, and D. Wei, "Multiscale CNN based on component analysis for SAR ATR," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2022, Art. no. 5211212.

[28] J. Zhang, M. Xing, G.-C. Sun, and Z. Bao, "Integrating the reconstructed scattering center feature maps with deep CNN feature maps for automatic SAR target recognition," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[29] J. Zhang, M. Xing, and Y. Xie, "FEC: A feature fusion framework for SAR target recognition based on electromagnetic scattering features and deep CNN features," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2174–2187, Mar. 2021.

[30] J. Liu, M. Xing, H. Yu, and G. Sun, "EFTL: Complex convolutional networks with electromagnetic feature transfer learning for SAR target recognition," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2021, Art. no. 5209811.

[31] M. J. Gerry, L. C. Potter, I. J. Gupta, and A. Van Der Merwe, "A parametric model for synthetic aperture radar measurements," *IEEE Trans. Antennas Propag.*, vol. 47, no. 7, pp. 1179–1188, Jul. 1999.

[32] K. Arulkumaran, M. P. Deisenroth, M. Brundage, and A. A. Bharath, "Deep reinforcement learning: A brief survey," *IEEE Signal Process. Mag.*, vol. 34, no. 6, pp. 26–38, Nov. 2017.

[33] V. Mnih et al., "Playing Atari with deep reinforcement learning," 2013, *arXiv:1312.5602*.

[34] Q. Liu et al., "A survey on deep reinforcement learning," *Chin. J. Comput. Sci.*, vol. 41, no. 1, pp. 1–27, 2018.

[35] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," 2017, *arXiv:1707.06347*.

[36] J. Hare, "Dealing with sparse rewards in reinforcement learning," 2019, *arXiv:1910.09281*.

[37] X. Wang, Y. Chen, and W. Zhu, "A survey on curriculum learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 9, pp. 4555–4576, Mar. 2021.

[38] L. C. Potter and R. L. Moses, "Attributed scattering centers for SAR ATR," *IEEE Trans. Image Process.*, vol. 6, no. 1, pp. 79–91, Jan. 1997.

[39] Y. Akyildiz and R. L. Moses, "Scattering center model for SAR imagery," *Proc. SPIE*, vol. 3869, pp. 76–85, Dec. 1999.

[40] Y. Akyildiz, "Feature extraction from synthetic aperture radar imagery," Ph.D. dissertation, Dept. Elect. Eng., Ohio State Univ., Columbus, OH, USA, 2000.

[41] M. A. Koets and R. L. Moses, "Feature extraction using attributed scattering center models on SAR imagery," *Proc. SPIE*, vol. 3721, pp. 104–115, Aug. 1999.

[42] J. R. Diemunsch and J. Wissinger, "Moving and stationary target acquisition and recognition (MSTAR) model-based automatic target recognition: Search technology for a robust ATR," *Proc. SPIE*, vol. 3370, pp. 481–492, Sep. 1998.

[43] H. Ling, R.-C. Chou, and S.-W. Lee, "Shooting and bouncing rays: Calculating the RCS of an arbitrarily shaped cavity," *IEEE Trans. Antennas Propag.*, vol. 37, no. 2, pp. 194–205, Feb. 1989.

[44] S. Feng, K. Ji, F. Wang, L. Zhang, X. Ma, and G. Kuang, "Electromagnetic scattering feature (ESF) module embedded network based on ASC model for robust and interpretable SAR ATR," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022, Art. no. 5235415.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:52:56 UTC from IEEE Xplore. Restrictions apply.